

Culture, Coordination, and Team Performance

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Abstract

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1 Introduction

Organizations increasingly rely on culturally diverse teams to solve complex tasks and compete in global markets. While diversity expands the range of knowledge, perspectives, and expertise available within teams, it may also create communication barriers and increase coordination costs. Consequently, whether cultural diversity ultimately enhances or hinders team performance remains a central question in management and economics. Economic theory emphasizes the trade-off between the informational benefits of diversity and the communication frictions arising from heterogeneous backgrounds (Lazear 1999), while organizational research similarly concludes that multicultural teams simultaneously benefit from broader informational resources but face greater communication difficulties and social integration challenges (Stahl et al. 2010). Organizational research further suggests that effective coordination depends not only on communication itself but also on the emergence of shared mental models—compatible understandings of the task, the environment, and teammates’ likely actions that allow individuals to anticipate one another without constant explicit communication (Cannon-Bowers et al. 1993; Klimoski and Mohammed 1994; Mathieu et al. 2000). Cultural similarity may facilitate the development of such shared understandings by reducing communication frictions and providing teammates with more compatible expectations about one another’s behaviour. Yet despite extensive evidence on how culture influences communication and

collaboration within organizations, comparatively little is known about whether cultural similarity ultimately translates into superior team performance.

In this paper, we study whether cultural similarity is associated with the performance of small, interdependent teams. We use professional men’s Grand Slam doubles tennis as an empirical model of organizational coordination, where two players with distinct individual abilities must repeatedly coordinate their actions to produce a joint outcome. Many organizations rely on dyadic or small-team collaboration, including project teams, surgical teams, military units, emergency response teams, and entrepreneurial partnerships. Although these settings differ substantially in purpose and context, they share a common organizational challenge: team members must repeatedly coordinate their actions under time constraints while adapting to rapidly changing environments. Doubles tennis captures these fundamental features in a particularly transparent way. Every point requires two teammates to coordinate positioning, movement, tactical decisions, and responses to their opponents, while success depends not only on individual ability but also on how effectively partners anticipate and complement each other’s actions. At the same time, performance is measured objectively, tasks are standardized across teams, and teammates operate under strong incentives to maximize joint success. These characteristics make doubles tennis an informative empirical model of organizational coordination in environments where communication and mutual understanding are essential to collective performance.

[rewrite](#) We collect data from all men’s doubles matches in Grand Slams between 2018 and 2025. We measure cultural similarity along three complementary dimensions: shared nationality, shared official language, and linguistic proximity. These measures capture different aspects of common communication and cultural background. We estimate whether culturally similar teams outperform culturally dissimilar teams in completed matches as well as in tiebreaks where outcomes are determined over a very small number of points, limiting the opportunity for coordination advantages to accumulate as a boundary condition.

We find that culturally similar teams are consistently more likely to win Grand Slam matches, with the strongest and most precisely estimated associations observed for the language-based measures of similarity. However, these advantages disappear in

tiebreaks. Rather than contradicting the coordination hypothesis, this pattern suggests an important boundary condition. Tiebreaks represent condensed episodes in which the competing teams have already demonstrated closely matched performance and relatively few points determine the outcome. Under these conditions, coordination advantages have less opportunity to accumulate and short-run variation plays a greater role in determining the winner. Finally, we find that the performance association of shared language and linguistic proximity becomes significantly stronger as teammates accumulate professional experience, consistent with communication advantages developing gradually through repeated interaction rather than arising automatically from common cultural backgrounds.

Our study relates to four strands of literature. First, it contributes to the literature on cultural diversity and organizational performance. Economic models emphasize the trade-off between diversity’s informational benefits and its communication costs (Lazear 1999), while organizational research documents that cultural diversity simultaneously enhances creativity and knowledge variety but also creates coordination challenges and interpersonal conflict (Stahl et al. 2010; Harrison and Klein 2007). Although numerous studies examine workforce diversity and firm outcomes, relatively few identify the role of cultural similarity within small teams performing highly interdependent tasks.

Second, our paper contributes to the literature on language, communication, and organizational coordination. Language has been recognized as a fundamental organizational resource that facilitates information exchange, knowledge integration, and collective decision-making (Wernerfelt 2004). In international business and organizational research, language differences have been shown to impede knowledge transfer and collaboration across multinational organizations (Marschan-Piekkari et al. 1999; Feely and Harzing 2003). Related evidence from economics shows that linguistic similarity reduces communication frictions and facilitates economic interaction, including trade, migration, and cross-border collaboration (e.g., Melitz and Toubal 2014). Our analysis extends this literature by examining whether shared language is associated with performance in teams whose success depends on continuous real-time coordination rather than one-off communication.

Third, we relate to the literature on shared mental models and team coordina-

tion. Organizational researchers argue that high-performing teams develop common understandings of tasks and teammates that enable implicit coordination, particularly in environments characterized by repeated interaction and rapidly changing conditions (Cannon-Bowers et al. 1993; Klimoski and Mohammed 1994; Mathieu et al. 2000). While this literature has primarily relied on laboratory experiments and organizational settings, our study provides evidence from elite professional teams performing under naturally occurring competitive conditions, where coordination quality is reflected directly in objective performance outcomes.

Finally, our paper builds on the emerging literature examining cultural homophily in professional sports. Most closely related is the recent Management Science study by Békés and Ottaviano (2025), which shows that professional football players exhibit systematic cultural homophily in passing behaviour, indicating that cultural similarity shapes patterns of collaboration even within highly professional organizations. Whereas that study focuses on who collaborates with whom, we examine whether cultural similarity is associated with how successfully teams perform once formed. By distinguishing collaboration from performance, we address a complementary question about the organizational consequences of cultural homophily.

This paper makes three contributions. First, it extends the emerging literature on cultural homophily by moving beyond collaboration patterns to examine objective team performance. Second, it contributes to the literature on communication and team coordination by showing that language-related dimensions of cultural similarity exhibit stronger and more persistent associations with performance than shared nationality alone, and that these associations become more pronounced as teams accumulate professional experience. Third, it identifies an important boundary condition for the value of cultural similarity. Rather than providing a universal performance advantage, cultural similarity appears to matter most in environments where teammates have repeated opportunities to coordinate over the course of a task and where communication advantages can accumulate over time, but much less in short, closely contested episodes such as tiebreaks, where outcomes are inherently more susceptible to point-level variation. More broadly, our findings suggest that cultural similarity is best understood not simply as a source of social affinity, but as a communication resource whose value depends on

the opportunities available for coordination to influence performance.

2 Data

2.1 Empirical Setting and Sample

Our empirical setting consists of men’s doubles matches from the four Grand Slam tournaments, the Australian Open, Roland Garros, Wimbledon, and the US Open, between 2018 and 2025. The dataset contains 1,876 completed Grand Slam matches.¹ Since each match contributes one winning and one losing team, the main analysis contains 3,752 units of team-match observations.

Player information is collected by combining official tournament records with ATP player information and supplementary public sources. Nationality and language information are fully harmonized through a multi-stage validation procedure that resolves inconsistencies arising from spelling variants, accented characters, historical nationality changes, and special cases such as Monaco. Only observations for which all four players have complete ranking and cultural information are retained.

2.2 Measuring Cultural Similarity

We examine three complementary measures of cultural similarity that capture different dimensions of communication and shared background.

The first measure, Same Nationality, equals one if both teammates represent the same country and zero otherwise. The second measure, Same Language, equals one if both teammates share an official national language. The third measure, Language Proximity, is a continuous measure ranging from zero to one that captures the linguistic similarity between teammates’ native languages, with larger values indicating greater linguistic proximity.

These measures reflect increasingly broad notions of cultural similarity. Shared nationality captures common institutional and cultural background. Shared language more

¹From the initial dataset with 1,933 matches we exclude retirements and walkovers because these outcomes do not reflect normal competitive performance. We further remove matches with incomplete doubles ranking information that cannot be reliably recovered, resulting in a final estimation sample of 1,876 matches.

Table 1: Observation Breakdown by Grand Slam and Year, and Descriptive Statistics

Panel A: Completed Matches by Tournament and Year					
Year	Australian Open	Roland Garros	Wimbledon	US Open	Total
2018	60	63	73	63	259
2019	62	60	61	60	243
2020	63	62	— ^a	31	156
2021	62	62	60	59	243
2022	59	62	59	62	242
2023	63	61	61	61	246
2024	60	58	61	63	242
2025	61	63	62	59	245
Total	490	491	437	458	1,876

Panel B: Descriptive Statistics (Team-Level, $N = 3,752$)				
Variable	Mean	SD	Min	Max
Team avg. doubles rank	126.5	183.9	1	1,395
Opponent avg. doubles rank	126.5	183.9	1	1,395
Years since turning pro (team avg.) ^b	11.9	5.2	1	27
Age at tournament (team avg.) ^c	31.6	4.4	16	43.5
Top-100 singles player (share)	0.081	0.272	0	1

^a Wimbledon 2020 was cancelled due to the COVID-19 pandemic.

^b *exp_mean*, i.e., tournament year minus year turned pro, averaged across the two teammates; imputed to 1 for 213 rookie team-obs whose raw tenure is ≤ 0 (turned pro in or after the tournament year).

^c *age_mean*, i.e., tournament year minus each player's birth year, averaged across the two teammates; built from date-of-birth data in the pipeline, with complete (zero-missing) coverage in this sample. Highly correlated with years since turning pro ($r = 0.83$); see Section 4.1 for why the two are used as alternative, not jointly-included, controls.

directly captures communication potential. Linguistic proximity recognizes that communication costs may also be lower between speakers of closely related languages even when they do not share the same official language.

Table 2: Shares of the Three Cultural Similarity Measures

Measure	Share of teams	N
Same nationality	41.2%	3,752
Same official language	56.2%	3,752
Language proximity (same value = 1) ^a	56.9%	3,752

^a In this sample *ling_prox* takes only the values 0 or 1 (1,616 vs. 2,136 team-obs); no team-pair in the estimation sample falls at an intermediate ethnolinguistic-proximity value.

Approximately 41% of teams consist of players sharing the same nationality, 56% share an official language, and 57% share a positive linguistic-proximity value.

3 Theory and Hypotheses

3.1 Cultural Similarity and Team Coordination

A central concept in the organizational literature is the idea of shared mental models. Teams develop shared representations of tasks, roles, priorities, and appropriate responses to different situations. When team members hold sufficiently similar understandings of these dimensions, they can anticipate one another’s actions and coordinate without having to communicate every decision explicitly. Related work on implicit coordination emphasizes that effective teams can rely on shared expectations when responding to recurring situations. Transactive memory systems provide a complementary perspective: through repeated interaction, team members develop an understanding of what their teammates know and how knowledge and responsibilities are distributed within the team.

These perspectives suggest that coordination depends partly on reducing the cognitive and communication costs of joint action. Culture and language can affect these costs. Teammates who share a language may communicate intentions more easily and interpret verbal and non-verbal cues more consistently. Shared nationality may capture additional dimensions of cultural familiarity, including norms, expectations, and com-

mon social experiences. More generally, linguistic proximity may reduce communication barriers even when teammates do not share a nationality or an official language.

The implication is not that cultural similarity directly makes individual players better. Rather, cultural similarity may make it easier for teammates to combine their individual abilities into effective joint performance. This distinction is central to our analysis. We are interested in the coordination return to cultural similarity, rather than in differences in individual ability associated with nationality, language, or culture. We therefore propose:

Hypothesis 1. *Teams with greater cultural similarity are more like to win matches than culturally dissimilar teams.*

3.2 Boundary Condition: Tiebreaks

The coordination benefits associated with cultural similarity should not necessarily be equally observable in every part of a match. Coordination is a process that can develop through repeated interaction: teammates communicate, adjust their behavior, learn one another's tendencies, and build shared expectations. A full match provides many opportunities for these processes to operate.

Tiebreaks provide a particularly informative boundary condition. [Add a short definition of tiebreaks here.](#) First, they are short relative to the match as a whole, leaving less time for a coordination advantage to accumulate. Second, a tiebreak is reached only when the two teams have performed sufficiently similarly to reach a closely contested stage of the set. Its outcome is therefore determined in a setting where relatively small differences in performance can be decisive and where point-level randomness is likely to play a larger role.

The communication environment also differs. During a set, teammates have repeated opportunities to communicate during changeovers, allowing them to exchange information, adjust tactics, and coordinate subsequent play. A tiebreak substantially compresses these opportunities. Once the tiebreak begins, teammates have very limited scope for formal discussion before the decisive points are played, with the principal changeover occurring only after the score reaches 6–6. Thus, a tiebreak provides not only a shorter horizon over which coordination can operate but also fewer opportunities

to communicate and update a shared understanding during that period.

These features make the tiebreak a useful test of the coordination interpretation. If cultural similarity improves team performance partly by facilitating communication, shared expectations, and coordinated action, its advantage should be less apparent in a short, closely contested sequence in which there are fewer opportunities for communication and greater scope for point-level randomness. This leads to our second hypothesis:

Hypothesis 2. *The performance advantage associated with cultural similarity is weaker in tiebreaks than in overall match outcomes.*

4 Empirical Analysis

4.1 Baseline Performance Effects

We begin by testing Hypothesis 1 by examining whether culturally similar teams outperform culturally dissimilar teams in Grand Slam doubles. Our baseline specification estimates the association between cultural similarity of the two players within a team and the probability of winning a match. Specifically, we estimate

$$\Pr(W_{ijt} = 1) = f(\beta C_{ijt} + \gamma X_{ijt} + \delta_{ty} + \lambda_r), \quad (1)$$

where W_{ijt} equals one if team i wins match j in tournament-year t , C denotes one of our three measures of cultural similarity: same nationality, same language, and language proximity.

The variables in X represent observable team and opponent qualities. Team quality is measured using the average ATP doubles ranking of the two teammates together with the average ranking of the opposing team. Lower ranking values indicate stronger players. We additionally include an indicator for whether either teammate has previously been ranked inside the ATP Top 100 in singles, capturing exceptional individual talent that may not be reflected fully in doubles rankings. Professional experience is measured as the average number of years since the two teammates turned professional. To allow for diminishing returns, we include both the demeaned experience measure and its squared term. Demeaning facilitates interpretation of interaction models while leaving the fitted

values unchanged.

Throughout the analysis we include tournament-by-year fixed effects δ_{ty} and round fixed effects λ_r to absorb common variation across tournaments and stages of competition. Standard errors are clustered at the match level.

Table 3: Match Win: Average Marginal Effects from Logit Models

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	0.0377** (0.0163)	0.0474*** (0.0156)	0.0434*** (0.0156)
Team avg. doubles rank	-0.0007*** (0.0001)	-0.0007*** (0.0001)	-0.0007*** (0.0001)
Opponent avg. doubles rank	0.0007*** (0.0001)	0.0007*** (0.0001)	0.0007*** (0.0001)
Top-100 singles player	-0.0049 (0.0315)	-0.0036 (0.0315)	-0.0030 (0.0315)
Team avg. yrs. since turning pro	0.0045*** (0.0017)	0.0043*** (0.0017)	0.0043** (0.0017)
Yrs. since turning pro, squared	-0.0008*** (0.0002)	-0.0007*** (0.0002)	-0.0007*** (0.0002)
Pseudo- R^2	0.095	0.096	0.096
N	3,752	3,752	3,752

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. FE: tournament $\times$ year, round. Outcome: match win (binary). Culture measures enter one at a time.

Table 3 reports average marginal effects (AMEs) estimated in logit models, which can be interpreted as percentage-point changes in the probability of winning associated with each explanatory variable.

Across all three specifications, culturally similar teams are significantly more likely to win Grand Slam matches. The estimated effects range from 3.8 to 4.7 percentage points and are strongest when cultural similarity is measured through shared language rather than shared nationality. Team quality remains the dominant predictor of success, as expected: stronger doubles rankings significantly increase the probability of winning, while more experienced teams also perform better, although the experience-performance relationship is concave.

These findings provide consistent evidence that cultural similarity is positively associated with team performance after accounting for observable differences in team quality, opponent strength, experience, and tournament characteristics.

4.2 A Boundary Condition: Tiebreak Performance

To test Hypothesis 2, we estimate the same empirical specification using individual tiebreaks as the unit of observation.

Table 4: Tiebreak Win: Average Marginal Effects from Logit Models

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	0.0298 (0.0223)	0.0075 (0.0218)	0.0019 (0.0217)
Team avg. doubles rank	−0.0003*** (0.0001)	−0.0003*** (0.0001)	−0.0003*** (0.0001)
Opponent avg. doubles rank	0.0003*** (0.0001)	0.0003*** (0.0001)	0.0003*** (0.0001)
Top-100 singles player	−0.0663 (0.0428)	−0.0634 (0.0427)	−0.0632 (0.0427)
Team avg. yrs. since turning pro	0.0026 (0.0022)	0.0022 (0.0021)	0.0022 (0.0021)
Yrs. since turning pro, squared	0.0000 (0.0003)	0.0001 (0.0003)	0.0001 (0.0003)
Pseudo- R^2	0.014	0.013	0.013
N	2,364	2,364	2,364

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. FE: tournament×year, round. Unit of observation: one team per (standard, 7-point) tiebreak, sets 1–5 (1,182 tiebreaks, 928 unique matches). Advantage-set deciders with no breaker played are not tiebreaks and are excluded.

Table 4 reports the corresponding tiebreak-win estimates. In contrast to the base-line analysis, none of the three measures of cultural similarity is significantly associated with the probability of winning a tiebreak. The estimated coefficients are economically small and statistically indistinguishable from zero. Importantly, the ranking controls remain highly significant, indicating that the specification continues to capture underlying differences in team quality.

Taken together, these findings suggest that the performance advantage associated with cultural similarity is not universal. Instead, it appears to emerge primarily over the course of an entire match, where teammates repeatedly coordinate across many points, rather than within short, closely contested episodes where relatively few points determine the outcome.

4.3 Robustness: Age at Tournament vs. Years Since Turning Pro

Team average player age at the tournament (*age_mean*, built from each player’s date of birth) is a second natural proxy for career maturity alongside years since turning

professional. The two are highly correlated: $\text{corr}(\text{age_mean}, \text{exp_mean}) = 0.83$, and a regression of demeaned age on demeaned experience and its square gives $R^2 = 0.70$, implying a variance inflation factor of 3.30 for age if entered alongside experience. Because entering both simultaneously destabilizes each tenure term’s own coefficient (Table 6 below), we follow standard practice for near-collinear regressors and estimate age and experience in separate specifications rather than jointly in the Table 3/4 main specs. Age coverage is complete (zero missing) in both the match-win and tiebreak-win samples, so unlike experience it requires no rookie-style imputation.

Table 5: Match Win: Age Spec (age_mean Replacing exp_mean)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	0.0355** (0.0164)	0.0467*** (0.0157)	0.0428*** (0.0157)
Team avg. age at tournament	0.0055** (0.0023)	0.0054** (0.0023)	0.0052** (0.0023)
Age, squared	−0.0006 (0.0004)	−0.0006 (0.0004)	−0.0006 (0.0004)
Pseudo- R^2	0.093	0.094	0.094
N	3,752	3,752	3,752

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Same sample, FE, and clustering as Table 3; rank and top-100 controls included but omitted for brevity (unchanged from Table 3). Culture AMEs are essentially unchanged from the experience-based main spec (e.g. same nationality: 0.0377 $\rightarrow$ 0.0355, both significant at 5%), confirming the baseline finding does not depend on which career-maturity proxy is used.

Table 6: Match Win: Age and Experience Entered Jointly (Diagnostic, Not a Preferred Spec)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture measure	0.0396**	0.0485***	0.0443***
Team avg. yrs. since turning pro	0.0025 (0.0029)	0.0025 (0.0029)	0.0026 (0.0029)
Team avg. age at tournament	0.0032 (0.0040)	0.0030 (0.0040)	0.0027 (0.0040)
N	3,752	3,752	3,752

Clustered (by match) standard errors in parentheses. ** $p < 0.05$, *** $p < 0.01$. With both tenure proxies in the model, neither the linear experience term nor the linear age term remains statistically significant ($p > 0.37$ for both, in every culture spec), and their standard errors roughly double relative to their single-proxy specs. This is the expected signature of near-collinear regressors, not evidence that career maturity stops mattering: Tables 3 and 5 each show a significant tenure effect on its own. The culture AMEs themselves remain robust and significant throughout.

The tiebreak-win null result (Table 4) is similarly robust to this choice: re-estimating with age_mean_dm and age_mean_dm² in place of the experience terms leaves none of the three culture measures significant (AMEs of 0.0243, 0.0036, and −0.0020 for same

nationality, same language, and language proximity respectively, all $p > 0.28$), and age itself is not significantly associated with tiebreak win ($p > 0.7$), mirroring experience’s own null association with tiebreak outcomes in Table 4.

4.4 Robustness check: Testing Tennis-Specific Explanations

One concern is that the observed relationship reflects strategic characteristics unique to particular playing surfaces rather than general coordination processes. Grass courts reward shorter rallies and faster exchanges, whereas clay courts involve longer rallies and different tactical patterns. If the observed association were driven by a particular tactical or environmental feature of doubles tennis, we might expect the return to cultural similarity to vary across surfaces, which differ substantially in their playing characteristics. We therefore test whether the association differs between grass and other surfaces and between clay and other surfaces.

We interact each cultural similarity measure with surface, estimating two specifications that each pool two surfaces into a single baseline rather than using all three simultaneously: grass vs. the rest (hardcourt + clay), and clay vs. the rest (hardcourt + grass). Because only two surface groups are used at a time, the surface main effect is separately identifiable from round fixed effects without tournament $\times$ year fixed effects, which would otherwise absorb it.

Table 7: Heterogeneity: Culture $\times$ Surface

Variable	(1) Same Nationality		(2) Same Language		(3) Lang. Proximity	
	(a) grass	(b) clay	(a) grass	(b) clay	(a) grass	(b) clay
Culture (C), pooled baseline	0.0392** (0.0183)	0.0374** (0.0188)	0.0438** (0.0176)	0.0498*** (0.0181)	0.0415** (0.0176)	0.0422** (0.0180)
Surface (grass in (a) / clay in (b))	0.0063 (0.0148)	−0.0007 (0.0142)	−0.0062 (0.0206)	0.0059 (0.0188)	−0.0022 (0.0207)	−0.0024 (0.0194)
$C \times$ Surface	−0.0082 (0.0373)	−0.0003 (0.0346)	0.0140 (0.0361)	−0.0106 (0.0340)	0.0075 (0.0362)	0.0038 (0.0344)
Pseudo- R^2	0.095	0.095	0.096	0.096	0.096	0.096
N	3,752		3,752		3,752	

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Controls: team/opponent avg. rank, top-100 singles, exp. (demeaned) and its square, as in Table 3. FE: round only (no tournament $\times$ year FE, so the surface main effect is separately identified). A robustness spec that restores tournament $\times$ year FE and instead identifies $C \times$ grass and $C \times$ clay jointly (hardcourt baseline, no separate surface main effect) agrees with the estimates above: both interactions remain small and insignificant ($p > 0.77$) in every culture measure.

The coordination advantage associated with cultural similarity is remarkably stable across playing environments: all surface main effects and Culture $\times$ Surface interactions

in Table 7 are statistically insignificant ($p > 0.67$ throughout). The result is not driven by any specific physical environment of the tournament.

5 Mechanisms

5.1 Coordination Through Shared Experience

On the one hand, experienced professionals have developed richer tactical knowledge and communication routines, allowing them to exploit shared language more effectively.

- Does cultural similarity become more valuable as teams accumulate experience?

We test this mechanism by augmenting the baseline specification by interacting each measure of cultural similarity with average years since turning professional.

We estimate the equation

$$\Pr(\text{Win}) = f(\beta_1 C + \beta_2 \text{AvgTeamExp} + \beta_3 (C \times \text{AvgTeamExp}) + \gamma X + \delta_{ty} + \lambda_r) \quad (2)$$

where AvgTeamExp is *exp_mean_dm*, the team-average years since turning professional, demeaned so that β_1 is evaluated at the sample-mean tenure (11.90 years) rather than at zero.

Table 8: Heterogeneity: Culture $\times$ Years Since Turning Pro

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C), mean experience	0.0377** (0.0163)	0.0450*** (0.0156)	0.0410*** (0.0156)
$C \times \text{exp_mean_dm}$	0.0003 (0.0032)	0.0064** (0.0032)	0.0063** (0.0032)
Pseudo- R^2	0.095	0.097	0.097
N	3,752	3,752	3,752

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Full Table 3 control set (team/opponent avg. rank, top-100 singles, *exp_mean_dm*, *exp_mean_dm_sq*) and FE (tournament $\times$ year, round) included but omitted from the table for brevity. A robustness spec adding the quadratic interaction $C \times \text{exp_mean_dm}^2$ leaves the linear interaction essentially unchanged and the quadratic term is itself insignificant throughout ($p > 0.50$).

Table 8 shows that the culture main effects at mean experience are close to the baseline Table 3 estimates. The interaction with team experience is positive and significant for the two language-based measures ($p \approx 0.045$ – 0.049): the performance association of shared language and linguistic proximity grows stronger as teams accumulate professional experience. Same-nationality homophily shows no comparable experience gradient

($p = 0.925$), consistent with communication advantages developing gradually through repeated interaction rather than arising automatically from shared nationality alone.

5.2 Knowledge Transfer with Experience Gaps

On the other hand, shared language becomes especially valuable when teammates differ substantially in expertise because communication and knowledge transfer become more demanding. Large experience gaps imply asymmetric knowledge, asymmetric decision authority, coaching/mentoring, and more need to transfer tacit knowledge. That is exactly where shared language may matter most.

- Does cultural similarity matter particularly when teammates differ substantially in experience?

The estimating equation becomes

$$\Pr(\text{Win}) = f(\beta_1 C + \beta_2 \text{TeamExpGap} + \beta_3 (C \times \text{TeamExpGap}) + \gamma X + \delta_{ty} + \lambda_r) \quad (3)$$

where TeamExpGap is *exp-gap-dm*, the (demeaned) absolute difference between the two teammates' own years since turning professional, $|\text{exp}_i - \text{exp}_j|$ — distinct from the team-average tenure used in Table 8 — and is included as an added control alongside its interaction.

Table 9: Heterogeneity: Culture $\times$ Within-Team Experience Gap

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C), mean gap	0.0373** (0.0163)	0.0462*** (0.0156)	0.0425*** (0.0156)
$C \times \text{exp_gap_dm}$	0.0032 (0.0033)	0.0038 (0.0032)	0.0036 (0.0032)
exp_gap_dm (main)	−0.0036* (0.0021)	−0.0045* (0.0025)	−0.0044* (0.0025)
Pseudo- R^2	0.096	0.097	0.096
N	3,752	3,752	3,752

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Full Table 3 control set and FE included but omitted for brevity. A robustness spec adding the quadratic interaction $C \times \text{exp_gap_dm}^2$ leaves this conclusion unchanged (quadratic term and interaction both insignificant, $p > 0.55$).

The interaction in Table 9 is positive throughout — the direction implied by the hypothesis that shared culture substitutes for an experience mismatch — but is not statistically significant for any of the three culture measures ($p = 0.24\text{--}0.34$). The

experience-gap main effect itself is negative and marginally significant ($p \approx 0.07$ – 0.08): teams with a larger tenure mismatch between teammates perform somewhat worse on average, independent of culture. This is a distinct, and so far null, result from Table 8’s finding that cultural advantage grows with a team’s own *average* experience.

5.3 Alternative Explanation: Individualism vs Collectiveness

Is the value of cultural similarity greater when the underlying cultural environment places more emphasis on collectivism? We augment the baseline specification with each team’s average Hofstede individualism score (IDV, demeaned; mean = 62.98, SD = 20.85) and its interaction with culture.

Table 10: Heterogeneity: Culture $\times$ Hofstede Individualism (Demeaned)

Variable	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C)	0.0306* (0.0166)	0.0401** (0.0162)	0.0360** (0.0162)
IC_team_dm	0.0011* (0.0006)	0.0008 (0.0008)	0.0008 (0.0008)
$C \times$ IC_team_dm (θ)	−0.00049 (0.00079)	−0.00003 (0.00089)	−0.00001 (0.00089)
Pseudo- R^2	0.096	0.097	0.097
N	3,752	3,752	3,752

Clustered (by match) standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Full Table 3 control set and FE included but omitted for brevity. A robustness spec dropping team-obs whose IDV score relies on a moderate-confidence regional proxy ($N = 3,597$) shows point estimates on C shrinking somewhat but not changing sign; the θ interaction remains small and insignificant throughout.

We find no evidence that individualism moderates the cultural homophily advantage: the interaction θ in Table 10 is small and statistically insignificant across all three culture measures ($p > 0.53$). The main IC_team_dm coefficient is positive and marginally significant for same nationality ($p = 0.052$) but insignificant for the language-based measures ($p > 0.27$).

6 Descriptive Evidence on Team Formation

6.1 Partner Selection

The regressions above estimate the association between cultural homophily and *performance*, conditional on a partnership having already formed. A separate concern is

selection: if stronger players have more choice over whom they partner with, and systematically choose culturally similar partners, that alone could produce the observed same-culture win advantage without cultural fit doing anything causal on court. This section tests whether partner selection is more culturally assortative among stronger players than among the field as a whole.

The unit of analysis is one row per distinct partnership per tournament (2,101 partnerships with complete ranking data across the four Grand Slams, 2018–2025; Olympics excluded). We compare the actual same-nationality, same-language, and mean language-proximity rates against a closed-form random-matching benchmark: for each tournament-year, the expected rate under random pairing of that event’s actual field, $\sum_{\text{pairs}} \mathbf{1}[\text{same}] / \binom{n}{2}$, computed from the same country-pair lookup underlying *same.language* and *ling.prox*.

Table 11: Partner Selection: Actual vs. Random-Matching Benchmark

Player pool	N	Same nationality		Same language		Mean ling.prox	
		Actual	Random	Actual	Random	Actual	Random
All ranked players	2,101	42.2%	5.2%	55.9%	16.2%	0.567	0.198
Both partners top-100	1,212	32.6%	4.1%	49.7%	15.5%	0.506	0.189
Both partners top-50	712	35.5%	4.5%	56.3%	15.8%	0.565	0.189

Random benchmark is a closed-form expectation under random pairing within each tournament-year’s actual field, not a simulation.

Two patterns stand out in Table 11. First, actual partner selection is far more culturally assortative than chance at every skill level — same-nationality rates run 6–8 $\times$ the random-matching benchmark, same-language rates 3–4 $\times$. Sorting by culture is real and substantial, and is worth keeping in mind when interpreting Tables 3–9 as associations between culture and winning rather than pure causal estimates of a coordination effect. Second, and less expected, elite players are not *more* assortative than the field as a whole: the actual-to-benchmark ratio for same nationality is essentially flat across brackets ($\approx 8.1\times$ overall, $\approx 7.9\times$ for top-100, $\approx 7.9\times$ for top-50). A complementary continuous test — regressing each player’s own ranking on the culture similarity of their realized partnership (4,202 ego-rows, clustered by partnership) — confirms this: the coefficient on own rank is positive and significant for all three culture measures ($p < 0.05$), meaning *worse*-ranked players sort into same-culture partnerships slightly *more*, not less. This is the opposite of what the “stronger players have more choice and sort more”

version of the selection concern predicts, and does not support selection as the sole driver of the performance associations documented above.

6.2 Evidence from the Olympics

If cultural similarity is associated with better performance, an important question is whether players appear to sort toward culturally similar partners when the incentives to coordinate are particularly salient. We therefore examine team composition around the Tokyo and Paris Olympic cycles.

This evidence is purely descriptive and uses only Grand Slam team-formation data, calendar-sliced around each Olympics; no Olympic-tournament observation enters the counts, table, or regression sample used anywhere else in this paper.²

Table 12: Team Composition Across Olympic Preparation Cycles (Grand Slams Only)

Cycle	Period	GS included	Same nationality	Same language	Ling. proximity
Tokyo 2021	Pre-Tokyo (Jan 2018–Dec 2019)	AO18–USO19	40.0%	54.8%	55.7%
	Tokyo Prep (Jan 2020–Jul 2021)	AO20–Wim21 ^a	40.3%	55.0%	55.1%
	Post-Tokyo (Aug 2021–Dec 2022)	USO21–USO22	39.0%	54.8%	55.6%
Paris 2024	Pre-Paris (Jan–Dec 2022)	AO22–USO22	39.3%	54.1%	55.2%
	Paris Prep (Jan 2023–Jul 2024)	AO23–Wim24	42.7%	57.9%	58.4%
	Post-Paris (Aug 2024–Dec 2025)	USO24–USO25	43.8%	58.8%	60.2%

^a Excludes Wimbledon 2020, cancelled due to the COVID-19 pandemic.

Shares of Grand Slam teams by calendar period relative to each Olympic Games; Olympic-tournament matches themselves are every row.

Table 12 reports the composition of teams across Olympic preparation cycles. Despite widespread speculation that Olympic eligibility may encourage players to form same-nationality partnerships, the share of same-nationality teams remains remarkably stable across the Tokyo Olympic cycle (40.0% pre-Tokyo, 40.3% during Tokyo prep, 39.0% post-Tokyo). The Tokyo preparation period coincides with the covid period. During that period, international travel and international interaction were unusually constrained.

If anything, this creates a plausible setting in which same-nationality partnering could have been especially attractive: easier travel; fewer international logistical complications; greater opportunities to train together; potentially greater value from exist-

²As a separate aside, doubles teams at the Olympic Games itself are required by IOC eligibility rules to share a nationality, so same-nationality, same-language, and ling.prox rates among the 126 Olympic team-obs in our data (62 in Tokyo 2021, 64 in Paris 2024) are mechanically 100% in every case — a reminder that the Olympics is categorically different from Grand Slam team formation and is why it is analyzed separately rather than pooled with the Grand Slam sample.

Cultural Homophily in Grand Slam Doubles - Tokyo 2021 Cycle
(Same vs. Diff, by Tournament-Year)

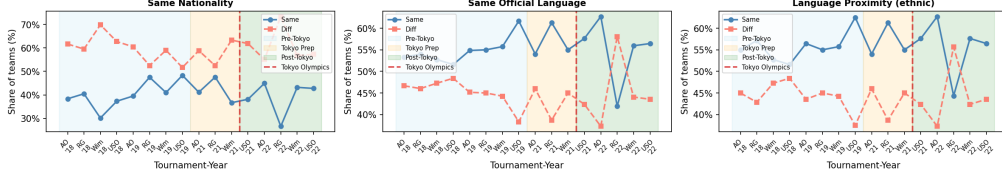


Figure 1: Cultural homophily in Grand Slam doubles across the Tokyo 2021 Olympic cycle, by tournament-year.

ing social/professional networks. Even in a period when practical constraints may have made same-nationality partnering particularly convenient, we do not observe an increase in same-nationality team formation (Figure 1).

During the Paris cycle, same-nationality and same-language partnerships become modestly more common (39.3% $\rightarrow$ 42.7% $\rightarrow$ 43.8% same-nationality; 54.1% $\rightarrow$ 57.9% $\rightarrow$ 58.8% same-language), although the increase occurs gradually across the whole 2022–2025 window rather than as a discrete shift during the Olympic preparation period specifically (Figure 2). These patterns do not provide evidence of systematic optimization around Olympic eligibility.

Cultural Homophily in Grand Slam Doubles - Paris 2024 Cycle
(Same vs. Diff, by Tournament-Year)

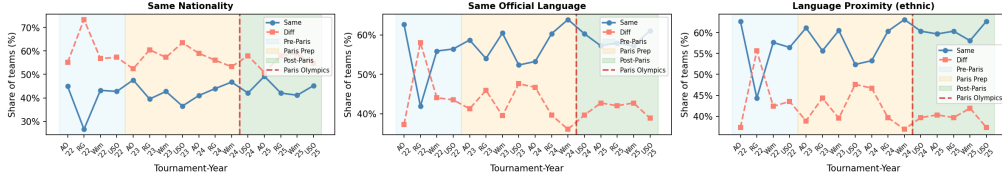


Figure 2: Cultural homophily in Grand Slam doubles across the Paris 2024 Olympic cycle, by tournament-year.

7 Conclusion

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